

Sameeksha Mahesh

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Education

University of California, Davis

Master of Science, Computer Science

Sep. 2025 – Mar. 2027 (Expected)

GPA: 4.0/4.0

- Relevant Coursework: Machine Learning & Discovery, Distributed Systems

Dayanand Sagar College of Engineering

Bachelor of Engineering, Computer Science & Engineering

Aug. 2020 – Jun. 2024

GPA: 3.97/4.0

Technical Skills

Languages	Java, C++, Python, SQL, JavaScript
Frameworks & Libs	Spring Boot, Flask, React, Node.js, Elasticsearch, Streamlit, dbt, Redis, Matplotlib, PyTorch
AI/ML	LLMs, Agentic AI, HuggingFace, Vector Embeddings, RAG Pipelines, LangChain, FAISS, TensorFlow
Developer Tools	Git, Docker, Kubernetes, Jenkins, GitHub Actions, Datadog, Postman, Linux
Databases	MySQL, PostgreSQL, MongoDB, Oracle, Snowflake
Cloud & DevOps	AWS (EC2, Lambda, EBS, S3), CI/CD Pipelines, Microservices

Experience

University of California, Davis

Jan. 2026 – Mar. 2026

Teaching Assistant — ECS 140A (Programming Languages)

- Led Java OOP and Haskell discussion sections and office hours for 100+ students, delivering targeted feedback on variables, scoping, exceptions, and debugging; designed an **automated code grader** in Python that eliminated manual grading for 100+ students, cutting evaluation turnaround from days to minutes and increasing submission iteration rate.

Perfios Software Solutions Pvt. Ltd

Apr. 2024 – Sep. 2025

Software Development Engineer

- Orchestrated 8+ microservices using **Docker** and **Kubernetes**; built end-to-end **CI/CD pipelines** (GitHub Actions + Jenkins) with staged rollouts, automated rollback, and integration testing, cutting deployment cycle time by **40%** and reducing production incidents from manual deploys.
- Built an intelligent document processing pipeline leveraging **Elasticsearch ML-based vector similarity** for bank identification, combining **OCR** text extraction with logo recognition to improve automated match rates from **25% to 65%** across international bank submissions.
- Shipped a full-stack invoicing platform from scratch (Spring Boot, JavaScript) with role-based access control (**RBAC**), AES-256 field-level encryption, and automated financial report generation; deployed to production, replacing a manual reporting workflow for enterprise clients.
- Redesigned RESTful microservices in Java and **Spring Boot** with caching, response compression, and async request handling, reducing API response times by **20%** across core payment services.

Brane Enterprises Pvt. Ltd

Jun. 2022 – Jan. 2023

Automation Intern

- Automated **20+ API** test workflows in **Postman** and integrated them into the **CI/CD** pipeline, reducing manual QA effort by **15%** and increasing test coverage by **25%**.
- Optimized 3 client-facing automation modules in **Node.js** across Agile sprint cycles, improving average task execution speed by **20%** with **Git**-managed version control.

Projects

Agentic RAG Document QA System | [Code](#)

April 2026

- Orchestrated a multi-step **agentic pipeline** coordinating chunking, **FAISS** retrieval, reranking, and LLM generation (**Ollama/OpenAI/LoRA**) with strict grounding and hallucination guardrails across multi-format document QA workloads.
- Engineered two-stage retrieve-and-rerank system (**bi-encoder** recall + **cross-encoder** reranking) that eliminated false-positive chunk retrieval, improving relevance scores by **~185%** on domain-specific queries — directly increasing answer precision for ambiguous multi-document inputs.

LocalBite – Community Food Delivery Platform | [Code](#)

Feb. 2026

- Architected a **full-stack** food delivery platform (React, Spring Boot, Docker/K8s) serving a 1,000+ user market; designed real-time operator dashboard with order tracking, dispatch state, and exception surfacing — cutting perceived commission overhead by **40–55%**.
- Designed a two-phase dispatch engine with **Redis** queues and **WebSockets**, cutting average dispatch time by **15%**; integrated **Stripe** split-payout settlement with full refund and cancellation support.